1.7 - Linear Independence

- **Def**: A set of vectors, $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is said to be <u>linearly independent</u> if the only solution to

"LI"

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n = \vec{0}$$

is $c_1 = c_2 = \dots = c_n = 0$. (the trivial solution).

A set of vectors is <u>linearly dependent</u> if it is not linearly independent. (so theres some other answer other than $\vec{0}$).

<u>Ex</u>: Are $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ LI?

so, solve $c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{array}\right] \xrightarrow[-3R_1 + R_3 \to R_3]{-2R_1 + R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & -6 & -6 & 0 \end{array}\right]$$

$$\xrightarrow{-2R_2 + R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

so, c_3 is free

$$\xrightarrow{\frac{-1}{3}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-4R_2 + R_1 \to R_1}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$c_1 - 2c_3 = 0 \longrightarrow c_1 = 2c_3$

$c_2 + c_3 = 0 \longrightarrow c_2 = -c_3$

so, $\vec{c} = \begin{bmatrix} 2t \\ -t \\ t \end{bmatrix}$ $t \in \mathbb{R}$

So, No! they are not LI b/c there's some other answer than $\vec{0}$.

* A set of one vector, $\{\vec{v}_1\}$ is LI as long as $\vec{v}_1 \neq \vec{0}$.

* A set of two vectors, $\{\vec{v}_1, \vec{v}_2\}$ is LI as long as neither vector is a multiple of the other.

<u>Ex</u>: $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$

are <u>Not</u> LI b/c $\vec{v}_2 = 4\vec{v}_1$

<u>Ex</u>: $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are LI

b/c neither $\vec{v}_1$ nor $\vec{v}_2$ is a multiple of $\vec{v}_2$ $(\vec{v}_1)$.

* A set of three OR more vectors is LI as long as there is no vector in the set that is a linear combination of other vectors in the set.

<u>Ex</u>: We know $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

are <u>NOT</u> LI, so one must be a linear combination of the other 2!

$\boxed{2\vec{v}_1 + \vec{v}_3 = \vec{v}_2}$ ← <u>Note</u>!

* If a set of vectors contains more vectors than the # of entries in each vector, then that set is not LI!

Ex: $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

are not LI as $3 > 2$ (# of vectors > # of entries)

→ In math-lingo:

$\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r\}$, $\vec{v}_i \in \mathbb{R}^n$, $1 \le i \le r$, is not LI if $r > n$.

→ As a matrix: If # columns > # rows, then the set of column vectors are not LI.

* If any of the vectors in a set of vectors is the zero vector, then the set is not LI!